

LESSON 6.1

PAYING YOUR WAY IN MATHLANDIA



LESSON INTRODUCTION

MA.7.AR.3.3 Solve mathematical and real-world problems involving the conversion of units across different measurement systems.

LEARNING TARGETS

- I can determine the exchange rate between two forms of currency.

BENCHMARK COHERENCE

BUILDING ON

MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates, and unit rates, including comparisons, mixtures, ratios of lengths, and conversions within the same measurement system.

WORKING TOWARD

MA.8.AR.3 Extend understanding of proportional relationships to two-variable linear equations.

ESSENTIAL QUESTIONS

- How are currency exchange rates used in different countries?
- How can you use the prices of goods to determine exchange rates?
- When comparing currency of different locations, what essential information is needed to calculate currency exchange rates?

LESSON NARRATIVE

In the lesson, students will begin developing their understanding of exchange rates between different forms of currency before formalizing it in Lesson 2. Students will engage in an Exploration through Mathlandia as stations to serve as different fictional places with a variety of items and currencies that students will navigate by identifying identical items with similar values to compare currency rates. Based on these comparisons, students will use ratios and rates to determine currency exchange rates. The concept of unit rate is applied but not formally introduced until Lesson 2.

COMPONENT SUMMARY

6.1.1 WARM-UP (~5-10 MIN.)

Explore ideas and examples of currency exchanges

6.1.2 EXPLORATION (20-25 MIN.)

Use given information about item prices to create currency unit rates

6.1.3 BRING IT TOGETHER (10 MIN.)

Practice writing ratios to represent the currency rates from the 6.1.2 Exploration

6.1.4 WRAP-UP (~5 MIN.)

Reflect on their learning after the lesson

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Exchange rates
- Reciprocal
- Unit Rate

MATERIALS

- Calculators
- (Optional) Timer
- (Optional) Items to represent the souvenir items

ADDITIONAL PREPARATION

- Video link for 6.1.1 Warm-Up
- Create stations in the classroom or otherwise determine how students will travel through the countries in Mathlandia for the 6.1.2 Exploration. Copies of Souvenir Menus should be available at each station.
- Determine how to group students for the 6.1.2 Exploration.
- The tables in 6.1.2 Exploration could be modified to represent fewer “lands” for students with disabilities.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may incorrectly use addition/subtraction reasoning. (i.e., “The magnet is 1.12 more sals than pops, so to change pops to sals, just add 1.12.”)
- Students may incorrectly reverse the divisor and dividend while dividing to find the exchange rates.
- Students may use the wrong exchange rate when finding prices of items in another currency, particularly in 6.1.2 Exploration.

THINGS TO CONSIDER

- Students should be allowed to use calculators so that the “messy” arithmetic does not hinder the conceptual understanding.
- If students get confused about unit rate and which number to divide, tell them to focus on the fact that the unit rate has to be per one of a certain unit.
- Ask students to consider that they may estimate exchange rates or write them as a range.

LITERACY AND LANGUAGE ROUTINES

Group Presentations

6.1.2 Exploration incorporates group presentations. Students are placed into groups to calculate currency rates for their “home countries” and organize the information on a given table. Data from all groups is collected on a common form and discussed by the entire class.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

6.1.2 Exploration provides an opportunity for students to engage in mathematical discussion by analyzing and critiquing methods to calculate unit rates for currency exchange. Use this as an opportunity for students to practice analyzing the thinking of others and then articulating their own thinking in response.

DIFFERENTIATE INSTRUCTION

Consider scaffolding instruction by breaking up the tables included in 6.1.2 Exploration and providing one or two answers. Students can discuss how the correct answer was derived to increase conceptual understanding.

The video in the 6.1.1 Warm-Up provides enrichment for auditory and visual learners. The souvenirs in 6.1.2 Exploration are common household items that can be displayed in the classroom to enhance the learning environment for auditory and visual learners.

The 6.1.4 Wrap-Up is an opportunity for students to reflect on the lesson learned during the lesson. For students who struggle with writing, consider allowing students to go back through the student workbook and place checks (understanding), question marks (need more clarification), and smaller statements next to specific activities.

6.1.1 WARM-UP: CURRENCY EXCHANGES

DIFFERENTIATE INSTRUCTION

Students may need background information about the euro. The teacher should encourage student discussion of the similarities and differences of both signs.

ENRICHMENT

How are currency rates related to the value of a country's form of currency? Why do you think currency rates are dynamic (changing) and not static (constant)?



6.1.1 Warm-Up: Currency Exchanges

If you travel to another country, chances are you'll need to exchange currency. For example, this could mean exchanging U.S. dollars for pesos of an equivalent value while visiting Mexico. Almost no two currencies are worth the same amount and their values change daily, sometimes even hourly. Currency exchange rates are used to determine the equivalent values at the time of exchange.



1. What are two things you wonder about currency exchanges? *Answers will vary.*
Is the U.S. dollar worth more or less than other similar sized counties?
Does the value of currency change dramatically in the same day?

An example of a currency exchange calculator is shown.

1	United States Dollar	=	0.8566	Euro
10	USD - United Sta...			
8.57	EUR - Euro			

2. What are two things you notice? *Answers will vary.*
The U.S. dollar is worth less than the Euro.
The calculator does not include a date or time, so it is not clear if this is current.
3. Peyton wrote the currency exchange as \$10 $\approx$ 8.57 Euros, €. The symbol, $\approx$, means "is approximately equal to." This is the symbol often used to show the relationship between currencies instead of an equal sign, =. *Why do you think that is? Answers will vary.*
The equal sign represents exact (absolute) amounts while the exchange rates represent approximated values at specific times. The equal sign should not be used for amounts that can change.



6.1.2 Exploration: Journey through Mathlandia

In class today, you will visit different countries in Mathlandia. In each country, you will see sets of souvenir items of the same value for sale, but with prices in each country's currency. Use the table to record the prices of the items in each country. Place a star beside your "home" country. As you visit each country, you will work together to determine the exchange rate in comparison to your "home country." Happy travels!

In the first column of the table, each country's name is listed in bold. The country's currency is listed below the country's name with the symbol for its units in parentheses. Write your "home" country's name and currency on the lines provided.

"Home" Country Name _____
 currency (units) _____
 Price of Goods by Country

	Mug	Keychain	Magnet	T-shirt	Picture Frame
Scalenia sal (\$)	6.27	4.73	3.15	16.86	8.24
Prismopolis peak (Å)	75.53	56.92	38	203.04	99.22
Proportia pop (P)	4.04	3.04	2.03	10.85	5.3
Equivaland equi (€)	4.75	3.58	2.39	12.77	6.24
Variableya var (v)	1.43	1.07	0.72	3.83	1.87
Integerina op (⊕)	36.81	27.75	18.52	98.97	48.36

6.1.2 EXPLORATION: JOURNEY THROUGH MATHLANDIA

DIFFERENTIATE INSTRUCTION

Students with disabilities, English language learners, or etc. may need modifications to this Exploration, as the amount of work required may be overwhelming. A modification would be to focus on the currencies from only two lands at a time. ELL and other students may find it helpful to highlight the currency unit or land. Additionally, consider students with experience living in countries with different currencies in your planning process. How could those students incorporate their own stories and perspectives to enrich their working knowledge of how to compare and contrast using ratios and rates?

TEACHER MOVES

As students work in groups during this Exploration activity, encourage students to participate in mathematical discussion while collecting and analyzing information used to calculate unit rates for currency exchanges. Within their home country, encourage students to listen and analyze others' mathematical language while determining the correct placement of pricing in the table.

6.1.2 EXPLORATION: JOURNEY THROUGH MATHLANDIA (CONTINUED)

ENRICHMENT

How are the terms “value” and “currency” the same? How are they different? Do items with different prices in different currencies have different values? Why or why not?

DIFFERENTIATE INSTRUCTION

To add complexity to Questions 2 and 3, have students research the exchange rate between the United States and the European Union from 2010 and 2020. Students can use an example of two common souvenirs to determine the actual exchange rate between the two currencies. Students can also discuss how the exchange rates have changed in ten years.

- Complete the statement to summarize the information given about the souvenir items in each country, using the words or phrases in the bank.

currency	equal to	value
----------	----------	-------

Each type of souvenir has the same value, which means it is worth the same amount of money. The price of each item is listed in that country's currency. Because the price of each souvenir from each country has the same value, the prices in different currencies can be set equal to each other. This can then be used to determine a currency exchange rate that can be used to approximate the cost of other items.

- With your partner, discuss how this information can be used to find the exchange rate between two currencies. Write a summary of your discussion.

Answers will vary.

To find the exchange rate between two currencies, you can start out with a common souvenir. Once you have the price of the souvenir for each unit of currency, set up a ratio. Divide the numerator and denominator of the ratio by the price for one of the currencies.

- Show an example of how to find the exchange rate between two currencies. Be sure to label the units.

Answers will vary.

$$\frac{4.75 \text{ €}}{4.04 \text{ ₪}} \div \frac{4.04}{4.04} \approx \frac{1.18 \text{ €}}{1 \text{ ₪}} \quad 1 \text{ ₪} = 1.18 \text{ €}$$

$$\frac{98.97 \text{ ₪}}{203.04 \text{ £}} \div \frac{203.04}{203.04} \approx \frac{0.49 \text{ ₪}}{1 \text{ £}} \quad 1 \text{ peak} = 0.49 \text{ op}$$

4. Use the information you gathered on your journey to calculate the exchange rates for each of the other currencies in comparison to your "home" currency. Record them in the section of the table for your "home" country's currency.

Round to the nearest hundredth.

Exchange Rate	Unit	Exchange Rate	Unit
0.64	pop(s)	1.55	sal(s)
12.05	peak(s)	18.70	peak(s)
0.76	equi(s)	1.18	equi(s)
0.23	var(s)	0.35	var(s)
5.87	op(s)	9.11	op(s)
Exchange Rate	Unit	Exchange Rate	Unit
0.08	sal(s)	1.32	sal(s)
0.05	pop(s)	0.85	pop(s)
0.06	equi(s)	15.90	peak(s)
0.02	var(s)	0.30	var(s)
0.49	op(s)	7.75	op(s)
Exchange Rate	Unit	Exchange Rate	Unit
4.38	sal(s)	0.17	sal(s)
2.83	pop(s)	0.11	pop(s)
52.82	peak(s)	2.05	peak(s)
3.32	equi(s)	0.13	equi(s)
25.74	op(s)	0.04	var(s)

5. Review the exchange rates displayed by other groups. Complete the table with the exchange rates for each currency.

6.1.2 EXPLORATION: JOURNEY THROUGH MATHLANDIA (CONTINUED)

DIFFERENTIATE INSTRUCTION

Consider providing chart paper and encourage students to represent the different currencies and their values in creative ways. For example, some students might find it best to draw currencies in different (smaller or larger) shapes as a pictorial representation of their values. Provide opportunities for learners with different entry points the ability to express their findings in different ways.

ENRICHMENT

What do you notice about the exchange rates from the peak to all of the other home rates?

Compare and contrast the equi and pop exchange rates.

TEACHER MOVES

Encourage students to participate in mathematical discussion while presenting activity findings to the entire class. Provide the suggestion that students presenting findings practice articulating their own thinking to the members of the home country prior to presenting to the entire class.

6.1.3 BRING IT TOGETHER: EXCHANGING CURRENCY

DIFFERENTIATE INSTRUCTION

Have students practice interpreting these mathematical statements, particularly reading the same statement in multiple ways. For example, 2.05 peaks equal 1 op, and 1 op represents approximately 2.05 peaks. In 8th grade this will become a discussion of constant rate of change or the slope on a graph.

TEACHER MOVES

Remind students to consider that when computing unit rates, it is important to complete tasks with mathematical fluency. The values calculated in exchange rates will not be exact values. However, the correct ratios must be used in order to ensure accurate rates.



6.1.3 Bring It Together: Exchanging Currency

1. Write two observations you made about the currencies from the activity.

Answers will vary.

The exchange rate from peak to the other currencies is very small.

The values for pop and equi appear to be closer to one another than any of the other rates.

2. Write a ratio that can be used to represent the exchange rate of ops ($\oplus$) to peaks ($\hat{\Lambda}$).

$$\frac{1 \text{ op}}{2.05 \text{ peaks}}$$

3. Write a ratio that can be used to represent the exchange rate of peaks ($\hat{\Lambda}$) to ops ($\oplus$).

$$\frac{1 \text{ peaks}}{0.49 \text{ ops}}$$

4. What do you notice about these two ratios?

Answers will vary.

One ratio is more than 1 (ops to peak) and the other is under 1 (peak to ops).

The ops currency is worth more than the peaks currency.

5. A hat costs 255 peaks ($\hat{\Lambda}$). Which ratio can be used to determine how much the same hat would cost in ops ($\oplus$)?

$$\frac{0.49 \text{ ops}}{1 \text{ peak}}$$

6. What is the cost of the hat in ops ($\oplus$)? Show your work.

$$255 \text{ peaks} \times \frac{0.49 \text{ ops}}{1 \text{ peaks}} = 124.95 \text{ ops}$$

**6.1.4 Wrap-Up: Reflection**

Complete each statement as you reflect on today's learning.

Answers will vary.

1. When someone asks me what I did in math class today, I can say ...
2. The thing that made the most sense to me today was ...
3. Something that I still do not understand is ...

6.1.4 WRAP-UP: REFLECTION**DIFFERENTIATE INSTRUCTION**

For students who might struggle representing their conceptual understanding through writing, consider alternative ways to assess or measure learning for this lesson. Students could choose to record themselves explaining the answer verbally, drawing and labeling, or providing an example. All forms can show an equal level of rigor and conceptual understanding.